UE23CS320A– Project Phase – 1
End Semester Assessment

Project Title : ***A Deep Learning and Generative AI-Driven Diet Recommendation System for Personalized Chronic Disease Management.***

Project ID : **17**

Project Guide : **Dr. Saritha**

Project Team : **Vatsal Jain** : PES2UG23CS681
Vivek Tutika : PES2UG23CS707
Pranav Shetty : PES2UG23CS721
Sourabh S Hegde : PES2UG23CS929

Problem Statement

- Existing systems offer one-size-fits-all nutrition suggestions.
- No continuous feedback loop for real-time dietary updates.
- Lack of patient trust and engagement with AI-only recommendations.
- No safety layer if the patient deviates from the suggested plan.

Abstract and Scope

Develop a deep learning–based system that provides real-time personalized diet plans for chronic disease management.
Integrate Generative AI to dynamically adjust diets based on patient input.

Scope:

- Focus on T2DM, CKD, and Hypertension.
- Include options for food preferences (veg, vegan, general preferences etc.).
- Offer supervision flexibility – family/doctor optional.
- Include an interactive chatbot for guidance.

Suggestions from Phase 2 - Review 1

Suggestions:

- User must not be asked to enter minute meal details. Needs to be more practical.
- Elaborate how the Risk Analysis module will be tackled.
- What hyperparameters should actually be defined?

Solutions:

- Users only log food items through search, barcode scan, or portion selection. Nutrients are computed automatically using verified nutrition datasets, and deviation detection is performed against the active diet plan.

- Clinical Risk Scoring Module: Aggregates user data, applies clinical guideline–based constraints (ADA, KDIGO, WHO), and uses ML models to compute disease-specific risk scores, combined into a final risk profile for alerts and diet adjustment.

Suggestions from Phase 2 - Review 1

Hyperparameters

Type Two Diabetes Mellitus (Random Forest Classifier)

- `n_estimators=120`
- `max_depth=4`
- `min_samples_leaf=125`
- `random_state=42`
- `n_jobs=-1`

Hypertension (Support Vector Classifier)

- `C=4`
- `kernel='rbf'`
- `probability=True`
- `random_state=42`

Chronic Kidney Disease (K-Nearest Neighbors Classifier)

- `n_neighbors=1`
- `weights='distance'`
- `algorithm='kd_tree'`

Suggestions from Phase 2 - Review 1

Clinical Guideline Sources:

- ADA (Diabetes – T2DM)
  - → https://diabetes.org
  - → https://diabetesjournals.org/care
- KDIGO (Chronic Kidney Disease)
  - → https://kdigo.org/guidelines/
- WHO (Hypertension / Sodium Intake)
  - → https://www.who.int/publications/i/item/9789241504836
  - → https://www.who.int/health-topics/nutrition
- ICMR-NIN (India Dietary Guidelines)
  - → https://www.nin.res.in

CKD Parameters

```
=== ckd_sample_1.json ===
{
  "disease": "ckd",
  "model": "KNeighborsClassifier",
  "predicted_class": 0,
  "raw_model_prediction": 0,
  "risk_score": 0.0,
  "risk_level": "low",
  "thresholds": {
    "moderate": 0.75,
    "high": 0.9
  }
}
```

```
### CKD Sample 1
  "age": 0.4772727272727273,
  "bp": 0.0769230769230769,
  "sg": 0.7500000000000071,
  "al": 0.0,
  "su": 0.0,
  "bgr": 0.155982905982906,
  "bu": 0.1388455538221528,
  "sc": 0.0013227513227513,
  "sod": 0.9175257731958762,
  "pot": 0.3333333333333335,
  "hemo": 0.8095238095238093,
  "pcv": 0.9111111111111112,
  "wbcc": 0.2272727272727273,
  "rbcc": 0.9545454545454544,
  "rbc": 1.0,
  "pc": 1.0,
  "pcc": 0.0,
  "ba": 0.0,
  "htn": 0.0,
  "dm": 1.0,
  "cad": 0.0,
  "appet": 0.0,
  "pe": 0.0,
  "ane": 0.0
```

Hypertention Parameters

Hypertension Sample 3
File: `demo_inputs/terminal_samples/hypertension_sample_3.json`

```json
{
  "male": 0.0,
  "age": 0.394736842105263,
  "education": 0.3333333333333333,
  "currentSmoker": 0.0,
  "cigsPerDay": 0.0,
  "BPMeds": 0.0,
  "prevalentStroke": 0.0,
  "diabetes": 0.0,
  "totChol": 0.2190669371196754,
  "sysBP": 0.7203647416413375,
  "diaBP": 0.8888888888888888,
  "BMI": 0.1199709161415414,
  "heartRate": 0.5612244897959184,
  "glucose": 0.0968660968660968,
  "TenYearCHD": 0.0
}
```

```
=== hypertension_sample_3.json ===
{
  "disease": "hypertension",
  "model": "SVC",
  "predicted_class": 1,
  "raw_model_prediction": 1,
  "risk_score": 1.0,
  "risk_level": "high",
  "thresholds": {
    "moderate": 0.488726,
    "high": 0.9
  }
}
}
```

T2DM Parameters

```
=== diabetes_sample_2.json ===
{
  "disease": "diabetes",
  "model": "RandomForestClassifier",
  "predicted_class": 1,
  "raw_model_prediction": 0,
  "risk_score": 0.460765,
  "risk_level": "moderate",
  "thresholds": {
    "moderate": 0.389735,
    "high": 0.661201
  }
}
```

Diabetes Sample 2

File: `demo_inputs/terminal_samples/diabetes_sample_2.json`

```json
{
  "Pregnancies": 0.3529411764705882,
  "Glucose": 0.4755244755244754,
  "BloodPressure": 0.4897959183673468,
  "SkinThickness": 0.258695652173913,
  "Insulin": 0.1465020576131687,
  "BMI": 0.1922290388548058,
  "DiabetesPedigreeFunction": 0.1265033407572383,
  "Age": 0.1333333333333333
}
```

Disease Scores

```
ne/Sem_6_Capstone"; & "d:/sem 6/capstone/.venv/Scripts/python.exe"
 "predict_from_terminal.py" --disease ckd --show-template
    "age": "<value>",
    "bp": "<value>",
    "sg": "<value>",
    "al": "<value>",
    "su": "<value>",
    "bgr": "<value>",
    "bu": "<value>",
    "sc": "<value>",
    "sod": "<value>",
    "pot": "<value>",
    "hemo": "<value>",
    "pcv": "<value>",
    "wbcc": "<value>",
    "rbcc": "<value>",
    "rbc": "<value>",
    "pc": "<value>",
    "pcc": "<value>",
    "ba": "<value>",
    "htn": "<value>",
    "dm": "<value>",
    "cad": "<value>",
    "appet": "<value>",
    "pe": "<value>",
    "ane": "<value>"
  }
```

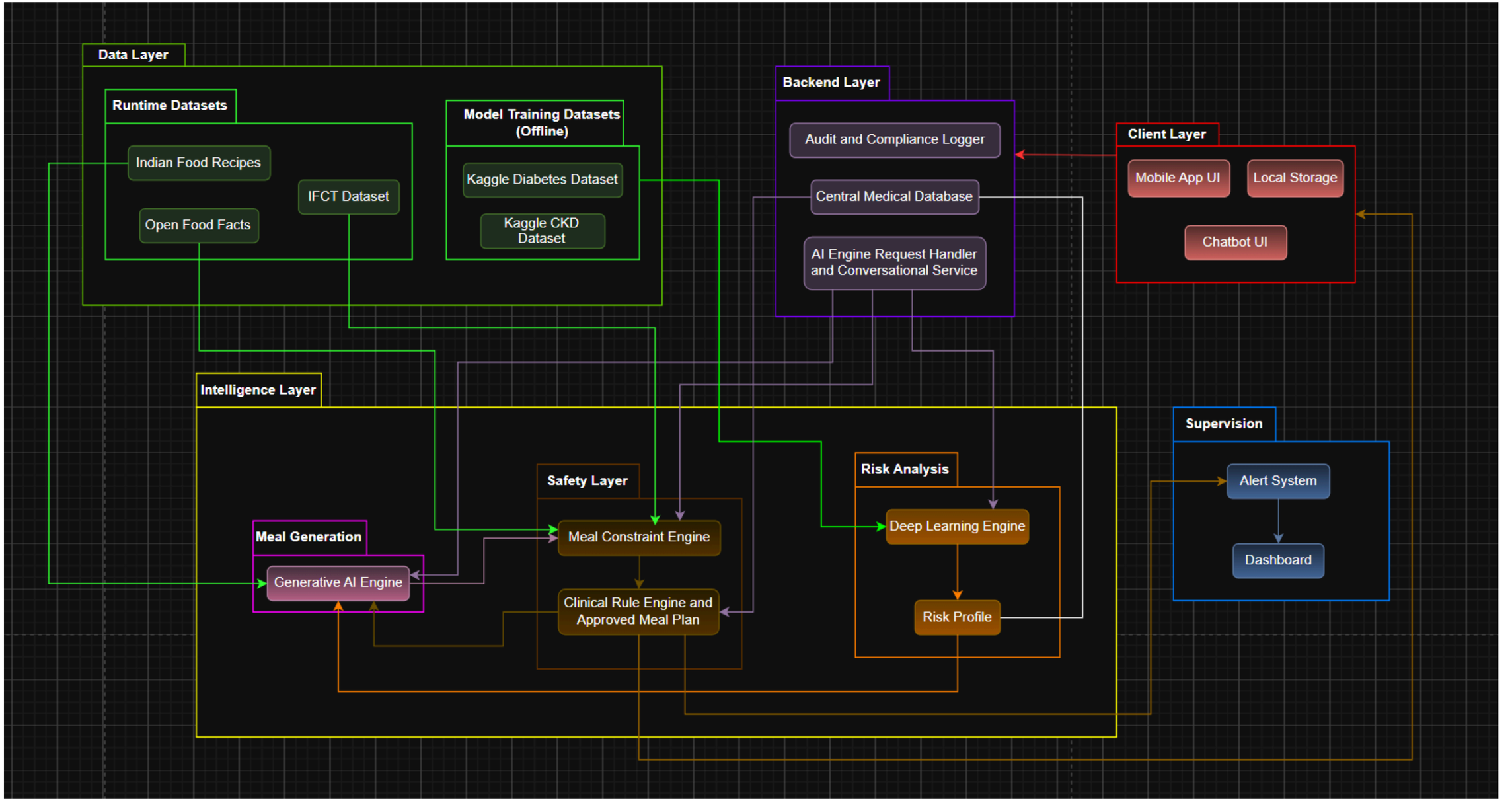
```
"predict_from_terminal.py" --disease hypertension --show-template

{
 "disease": "hypertension",
 "features": {
   "male": "<value>",
   "age": "<value>",
   "education": "<value>",
   "currentSmoker": "<value>",
   "cigsPerDay": "<value>",
   "BPMeds": "<value>",
   "prevalentStroke": "<value>",
   "diabetes": "<value>",
   "totChol": "<value>",
   "sysBP": "<value>",
   "diaBP": "<value>",
   "BMI": "<value>",
   "heartRate": "<value>",
   "glucose": "<value>",
   "TenYearCHD": "<value>"
 }
```

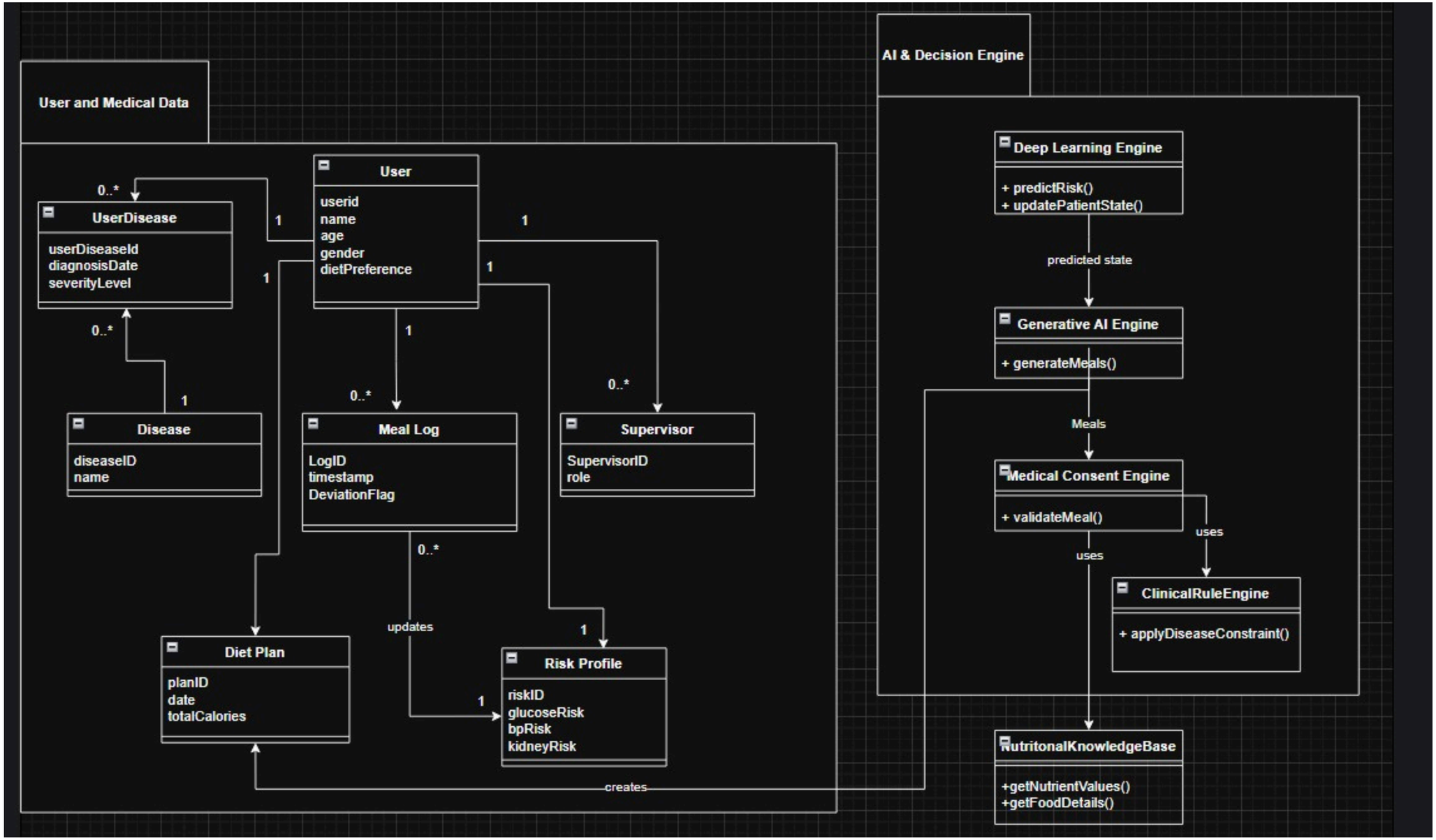
Disease Scores

```
ne/Sem_6_Capstone"; & "d:/sem 6/capstone/.venv/Scripts/python.exe"
 "predict_from_terminal.py" --disease diabetes --show-template
{
  "disease": "diabetes",
  "features": {
    "Pregnancies": "<value>",
    "Glucose": "<value>",
    "BloodPressure": "<value>",
    "SkinThickness": "<value>",
    "Insulin": "<value>",
    "BMI": "<value>",
    "DiabetesPedigreeFunction": "<value>",
    "Age": "<value>"
  }
}
```

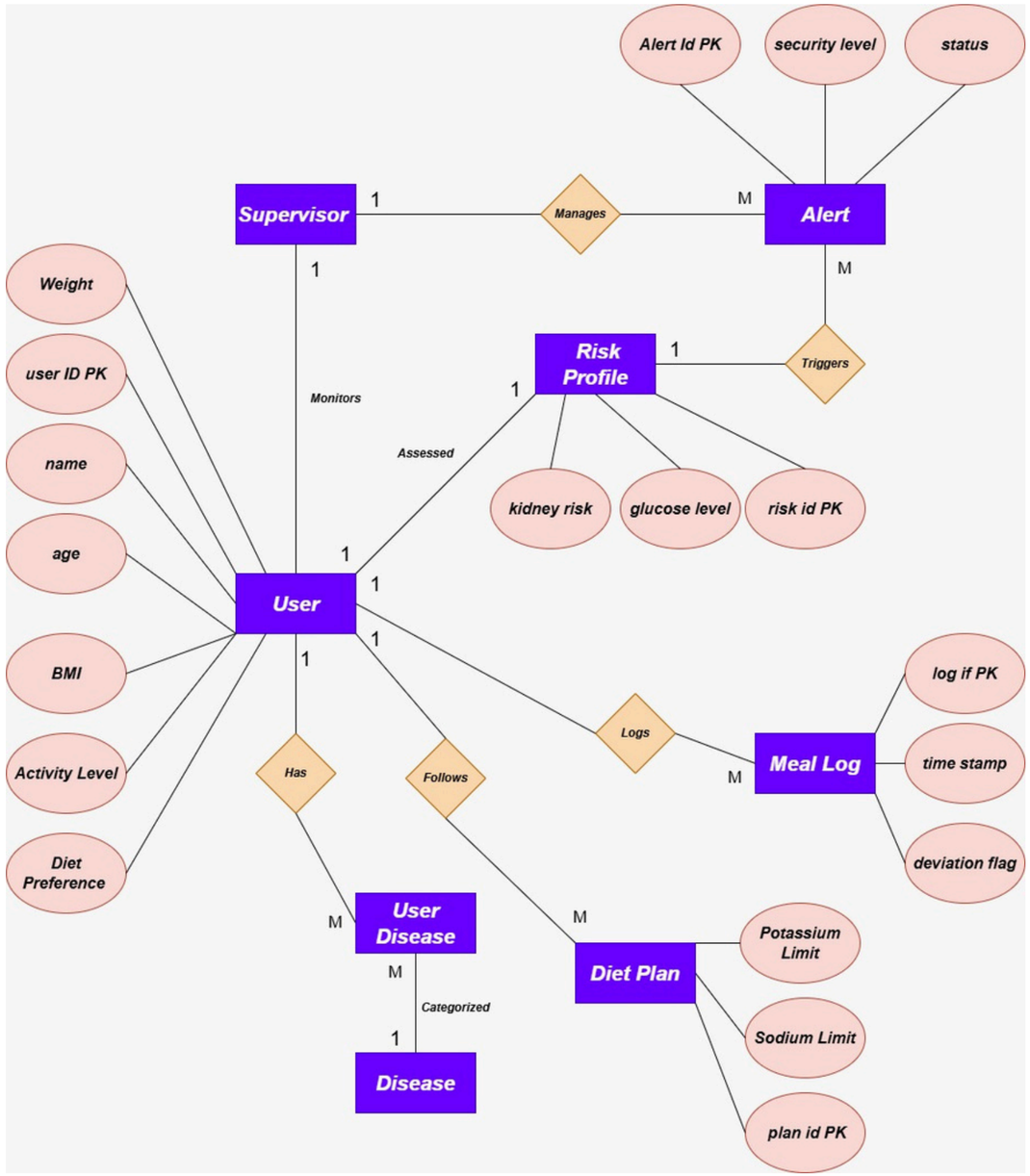
Architecture Diagram



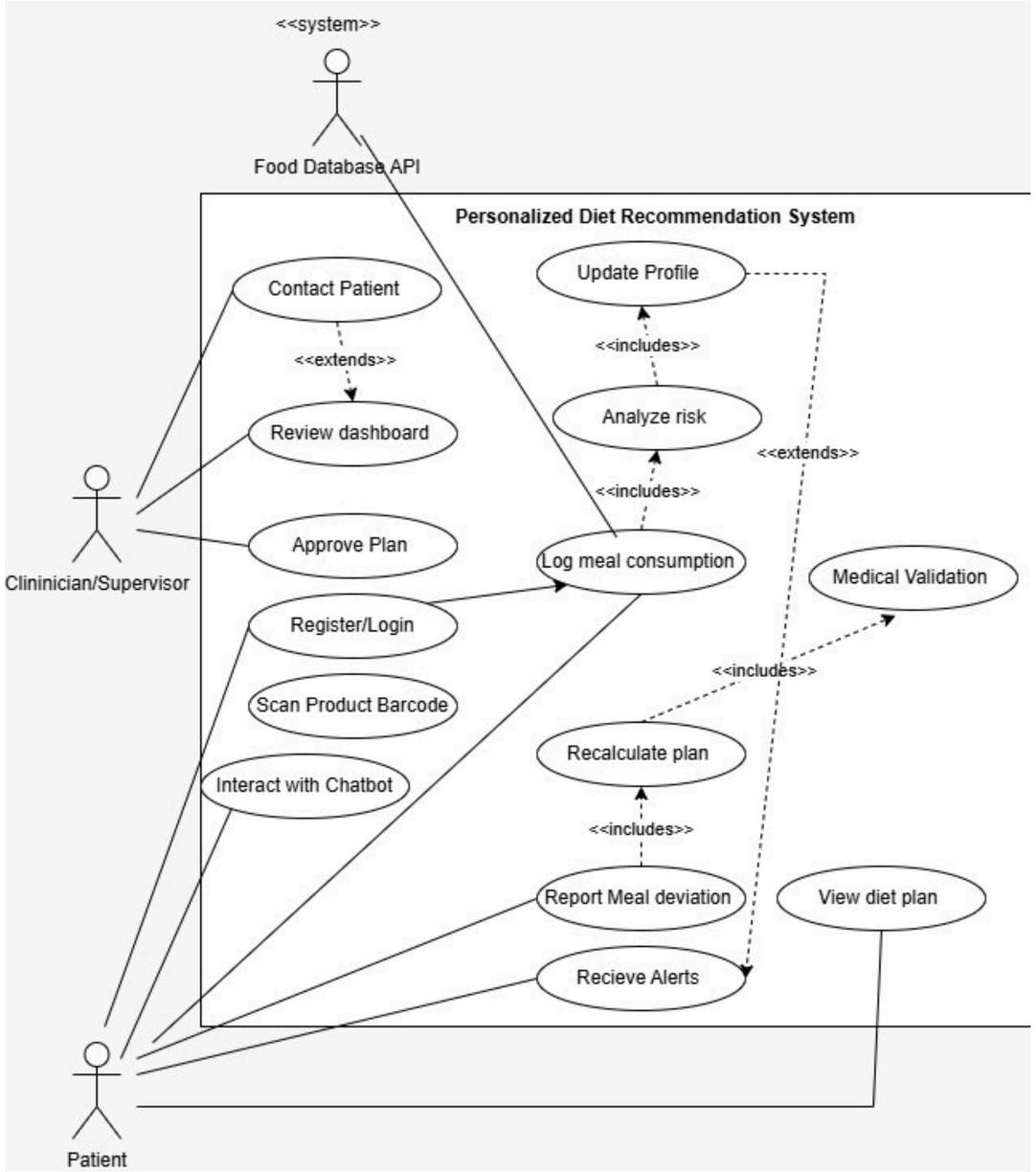
Master Class Diagram



ER Diagram


Alert Id PK
security level
status
Supervisor
1
Manages
M
Alert
1
M
Weight
user ID PK
name
age
BMI
Activity Level
Diet Preference
Monitors
Risk Profile
1
Triggers
1
Assessed
kidney risk
glucose level
risk id PK
1
User
1
1
1
Has
Follows
Logs
M
Meal Log
log if PK
time stamp
deviation flag
M
User Disease
M
Categorized
1
Disease
M
Diet Plan
Potassium Limit
Sodium Limit
plan id PK

Use-Case Diagram



Requirement Specification

1) Functional Requirements:

→ User meal logging with deviation detection (manual / assisted).
→ Real-time diet recalculation after deviation.
→ Disease-specific dietary enforcement (T2DM, CKD, Hypertension).
→ Generative AI–based adaptive meal suggestions.
→ Optional clinician/family supervision and alerting.
→ Interactive chatbot for dietary guidance.

2) Non-Functional Requirements:

→ Data privacy and anonymization.
→ Low-latency response for real-time recalculation.
→ High explainability for medical safety decisions.
→ Modular architecture for future dataset replacement.
→ Scalability to multi-disease patients.

Confirmed Datasets

1) Open Food Facts (India Subset)

- 50,000+ packaged food products available in India.
- Nutrition labels, allergens, additives, Nutri-Score.
- Open Database License (ODbL).
- Role in system:
  - → Processed and packaged food handling.
  - → Barcode-based food logging.
  - → Nutrient validation for real-world consumption.

2) Indian Food Composition Tables (IFCT) 2017 - ICMR-NIN

- 528 commonly consumed Indian food items.
- ~151 nutritional parameters per item.
- Macronutrients, micronutrients, vitamins, minerals.
- Region-specific Indian dietary representation.
- Role in system:
  - → Ground-truth nutrition reference.
  - → Medical constraint enforcement.
  - → Nutrient calculation for all meal plans.

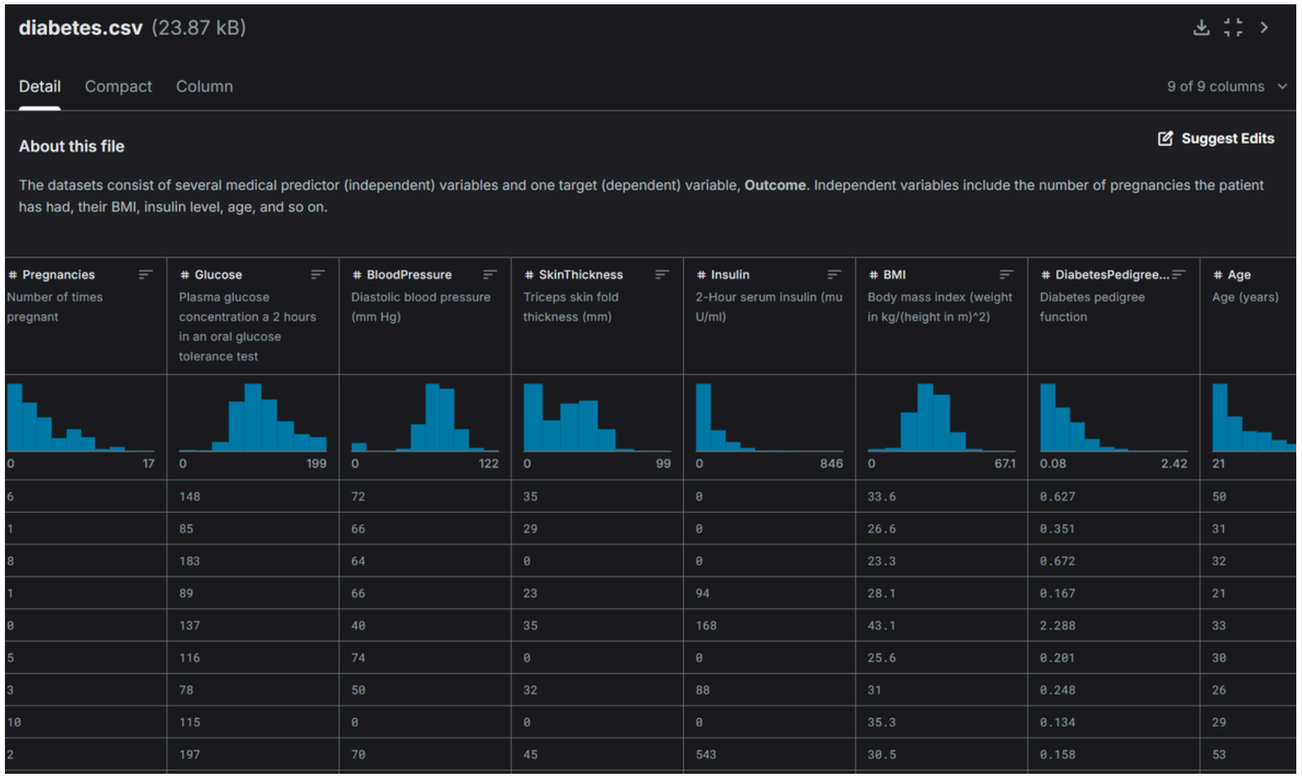
3) Indian Food Recipes Dataset – Mendeley Data

- ~6000 Indian dishes and recipes.
- Fields: ingredients, cooking method, region, diet type, nutrition estimates.
- Ensuring cultural and regional relevance in recommendations.
- Supporting diet-type filtering (veg / vegan / regional preferences).

Temporary Datasets

4) Kaggle Diabetes Datasets

- Files: diabetes.csv, diabetes_prediction_dataset.csv
- Features: glucose, BMI, age, insulin, HbA1c proxy, outcome.
- Usage:
  - → Initial risk classification models.
  - → Feature engineering validation.
- Limitation:
  - → No diet history.
  - → No longitudinal tracking.



diabetes.csv (23.87 kB)

Detail Compact Column 9 of 9 columns

About this file Suggest Edits

The datasets consist of several medical predictor (independent) variables and one target (dependent) variable, **Outcome**. Independent variables include the number of pregnancies the patient has had, their BMI, insulin level, age, and so on.

# Pregnancies Number of times pregnant	# Glucose Plasma glucose concentration a 2 hours in an oral glucose tolerance test	# BloodPressure Diastolic blood pressure (mm Hg)	# SkinThickness Triceps skin fold thickness (mm)	# Insulin 2-Hour serum insulin (mu U/ml)	# BMI Body mass index (weight in kg/(height in m)^2)	# DiabetesPedigree... Diabetes pedigree function	# Age Age (years)
0 – 17	0 – 199	0 – 122	0 – 99	0 – 846	0 – 67.1	0.08 – 2.42	21
6	148	72	35	0	33.6	0.627	50
1	85	66	29	0	26.6	0.351	31
8	183	64	0	0	23.3	0.672	32
1	89	66	23	94	28.1	0.167	21
0	137	40	35	168	43.1	2.288	33
5	116	74	0	0	25.6	0.201	30
3	78	50	32	88	31	0.248	26
10	115	0	0	0	35.3	0.134	29
2	197	70	45	543	30.5	0.158	53

Temporary Datasets

5) Kaggle Chronic Kidney Disease Dataset

- Files: kidney_disease.csv, kidney_disease_train.csv, kidney_disease_test.csv
- Features: creatinine, urea, sodium, potassium, hemoglobin.
- Usage:
  - → CKD risk and stage estimation.
  - → Constraint threshold validation.
- Limitation:
  - → Static snapshots.
  - → Synthetic imbalance and missing values.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1
5	5	116	74	0	0	25.6	0.201	30	0
6	3	78	50	32	88	31.0	0.248	26	1
7	10	115	0	0	0	35.3	0.134	29	0
8	2	197	70	45	543	30.5	0.158	53	1
9	8	125	96	0	0	0.0	0.232	54	1

Temporary Datasets

kidney_disease_test.csv (14.19 kB)

Detail Compact Column 10 of 25 columns

About this file Suggest Edits

Testing data for the challenge.

id	age	bp	sg	al	su	rbc	pc
0 – 396	5 – 82	50 – 110	1 – 1.02	0 – 4	0 – 5	normal 48% [null] 38% Other (18) 15%	normal abnormal Other (15)
209	19.0	70.0	1.02	0.0	0.0		normal
280	47.0	80.0					
33	60.0	100.0	1.02	2.0	0.0	abnormal	abnormal
210	59.0	100.0	1.015	4.0	2.0	normal	normal
93	73.0	100.0	1.01	3.0	2.0	abnormal	abnormal
84	59.0	70.0	1.01	3.0	0.0	normal	abnormal
329	33.0	60.0	1.025	0.0	0.0	normal	normal
94	65.0	70.0	1.01	0.0	0.0		normal
266	55.0	80.0	1.02	0.0	0.0	normal	normal
126	70.0	90.0	1.015	0.0	0.0		normal
9	53.0	90.0	1.02	2.0	0.0	abnormal	abnormal
361	29.0	80.0	1.02	0.0	0.0	normal	normal
56	76.0	70.0	1.015	3.0	4.0	normal	abnormal

kidney_disease.csv (48.55 kB)

Detail Compact Column 10 of 26 columns

About this file Suggest Edits

This is csv file. You can either go to the link above or use this file

id	age	bp	sg	al	su	rbc	pc
0 – 399	2 – 90	50 – 180	1 – 1.02	0 – 5	0 – 5	normal 50% [null] 38% Other (47) 12%	normal abnormal Other (65)
0	48.0	80.0	1.02	1.0	0.0		normal
1	7.0	50.0	1.02	4.0	0.0		normal
2	62.0	80.0	1.01	2.0	3.0	normal	normal
3	48.0	70.0	1.005	4.0	0.0	normal	abnormal
4	51.0	80.0	1.01	2.0	0.0	normal	normal
5	60.0	90.0	1.015	3.0	0.0		
6	68.0	70.0	1.01	0.0	0.0		normal
7	24.0		1.015	2.0	4.0	normal	abnormal
8	52.0	100.0	1.015	3.0	0.0	normal	abnormal
9	53.0	90.0	1.02	2.0	0.0	abnormal	abnormal
10	50.0	60.0	1.01	2.0	4.0		abnormal
11	63.0	70.0	1.01	3.0	0.0	abnormal	abnormal

Thank You